

Assembly + Use

Printing:

1. Print pages 1 and 2 at 100% / Actual Size. Verify either scale bar reads 100 mm.
2. Cut along the dashed seam line on each page (the inner edge of the marker).
3. Lay them on a flat surface, line up the registration crosses, and tape on the BACK side.
4. Measure the assembled circle with calipers across two perpendicular axes. Use the average.

Mounting:

Stick FLAT on the drone belly. Curling = ellipse = bad area math = bad Z.

If your drone is small, the 6" marker may overhang. The marker doesn't need to be centered on the drone, just FLAT and rigid. Ziptie a backing card if needed.

Firmware setup (cam_test_v3):

```
D152.4          # marker diameter in mm (use your measured value)
[aim cam at lit scene, wait ~2 sec for AE to settle]
X              # LOCK exposure + gain
[hold marker at known distance, e.g. 1000 mm]
C1000          # back-solve focal length
```

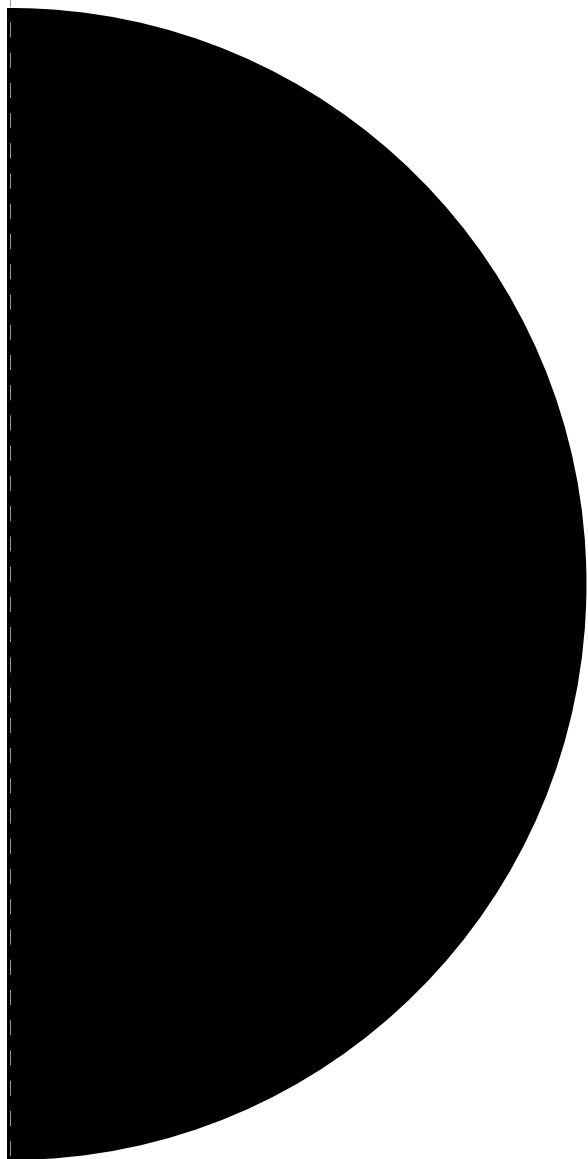
Why a bigger marker:

Z error from blob-area distance is roughly proportional to $1/\text{diameter_pixels}$.

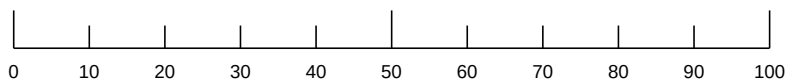
A 6" marker is ~3x bigger in the image than the 2" one, so Z is ~3x more accurate at every distance. Combined with SVGA frame size (2.5x more pixels per inch), the practical accuracy gain is roughly 7-8x vs the original setup.

6 inch (152.4 mm) marker — page 1 of 2 [left half]

Print at 100% / Actual Size. Both pages then align registration crosses and tape.

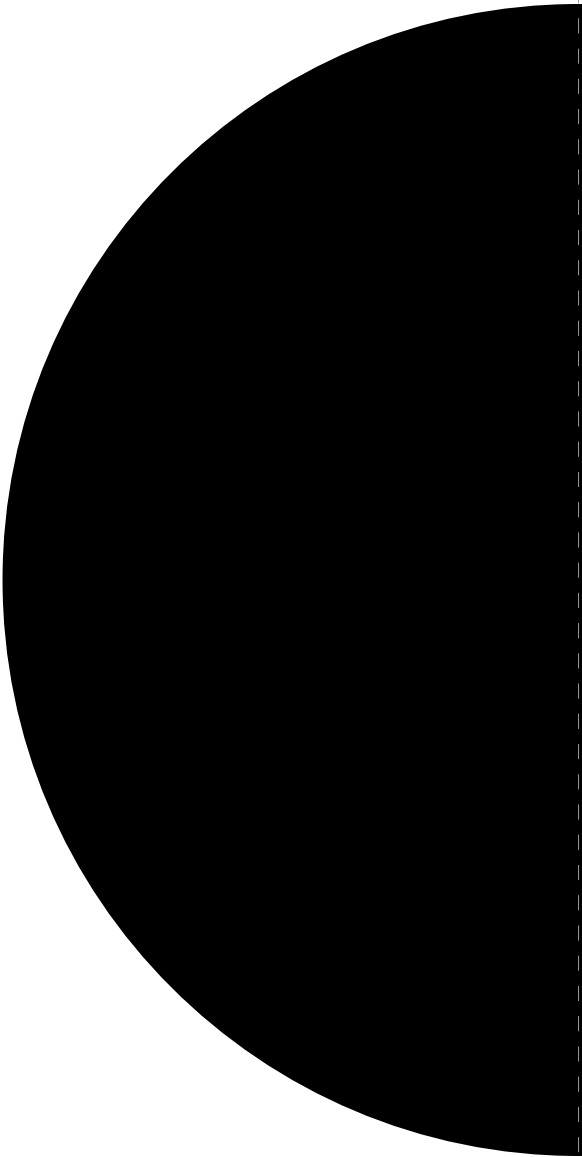


Scale: 100 mm



6 inch (152.4 mm) marker — page 2 of 2 [right half]

Print at 100% / Actual Size. Both pages then align registration crosses and tape.



Scale: 100 mm

